



# AUTONOMOUS RACING CAR

*Smart Sensors | Intelligent Control | Faster Tomorrow*

## **A 3D-Printed ESP32-S3 Autonomous Racing Car**

Design, Build and Programming Documentation

**Project Name : Autonomous Racing Car**

**Main Controller : ESP32-S3**

Course : Robotics & Embedded Systems

Year : 2026

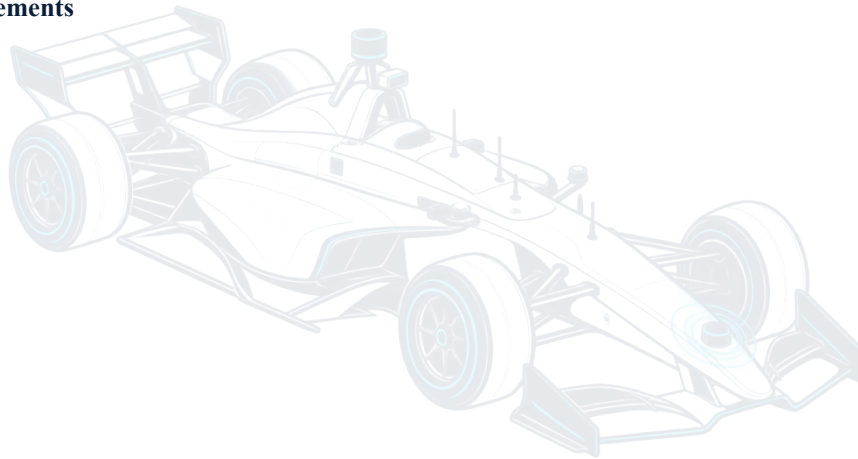




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# 1. Project Overview

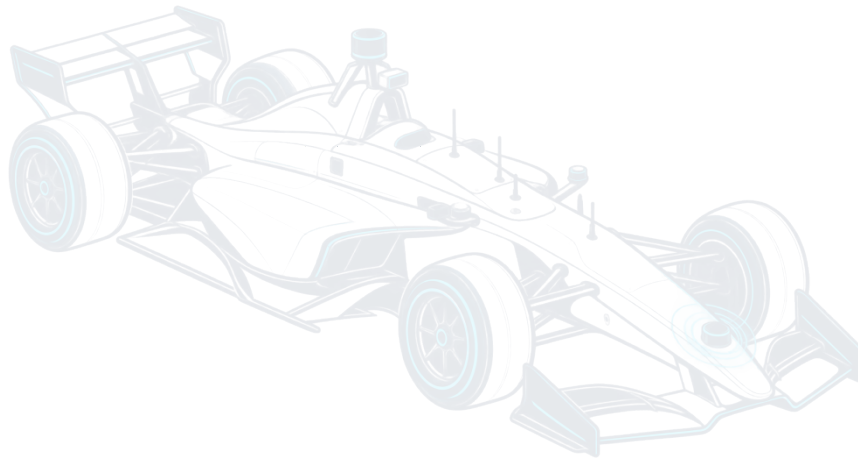
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The Autonomous Racing Car is an intelligent robotic vehicle designed to navigate a race track without continuous human control. It uses an ESP32-S3 as the main controller to process sensor information, control the motors, detect obstacles, calculate distances, and make decisions while racing.

The system combines a 360° LiDAR, wheel encoders, IMU, motor driver, steering servo and other sensors to allow the car to navigate safely and efficiently.

## Key Idea

*“Smart technology, precision control, maximum performance.”*

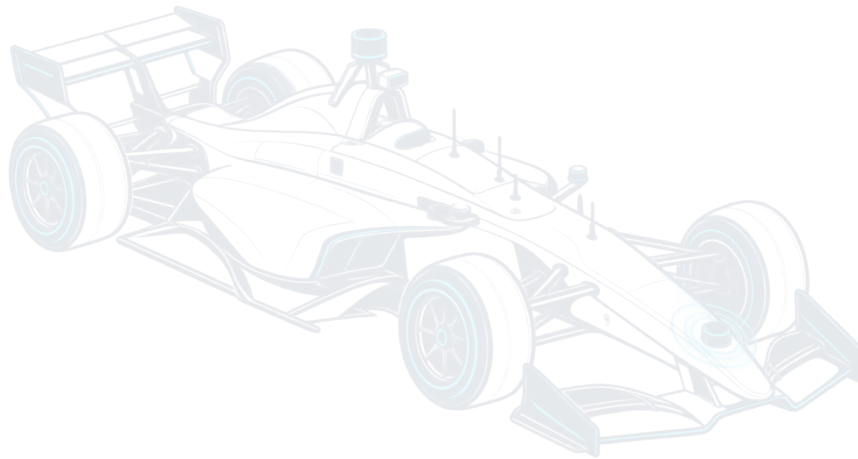




## 2. Objectives

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- Build a fully autonomous racing car.
- Use a 360° LiDAR to detect obstacles and other racing cars.
- Measure the distance between the car and obstacles.
- Calculate the distance traveled by the car using wheel encoders.
- Determine when the car needs to slow down before a turn.
- Automatically control steering and motor speed.
- Detect and avoid collisions.
- Navigate different turns on the racing track.
- Use the ESP32-S3 to control the complete system.
- Develop a reliable and responsive autonomous control system.

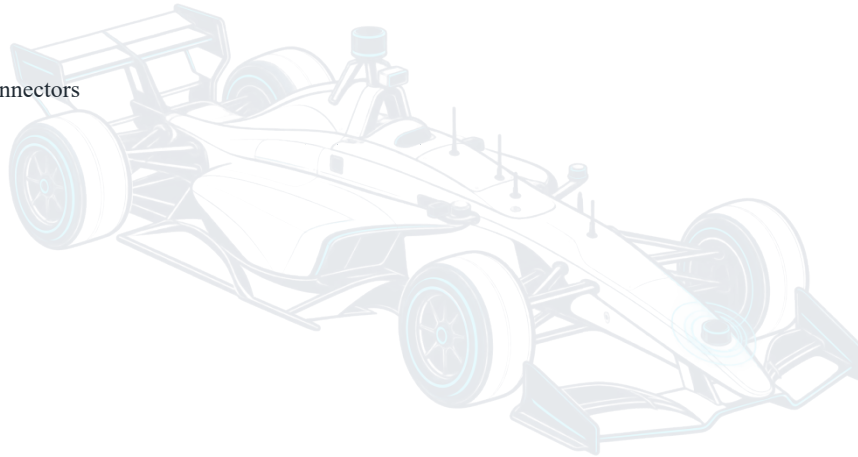




### 3. Hardware Components

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- ESP32-S3 Development Board
- 360° LiDAR Sensor
- Motor Driver (L298N)
- DC Geared Motor (6V)
- Wheel Encoder
- IMU Sensor (MPU6050)
- Steering Servo Motor
- Ultrasonic Sensor (HC-SR04)
- Emergency-Stop Button
- Power Button
- Rechargeable Battery (7.4V/11.1V)
- Buck Converter (5V/3A)
- Wheels (x4)
- 3D-Printed Chassis
- Jumper Wires & Connectors





## COMPONENT COSTING

### 4. Bill of Materials

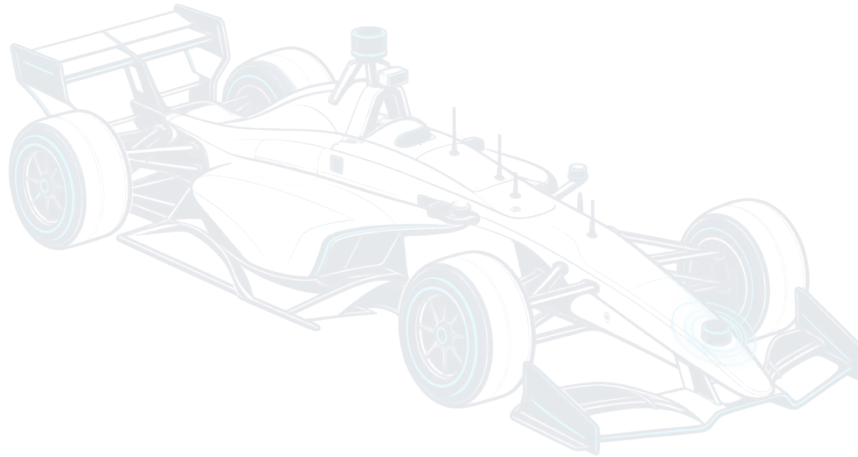
| No.         | Component            | Part Number / Model | Supplier      | Qty   | Unit Price (ZAR) | Total (ZAR) |
|-------------|----------------------|---------------------|---------------|-------|------------------|-------------|
| 1           | ESP32-S3 Dev Board   | ESP32-S3-WROOM-1    | Seeed Studio  | 1     | R 250            | R 250       |
| 2           | 360° LiDAR           | RPLIDAR A1          | RobotShop     | 1     | R 1,800          | R 1,800     |
| 3           | Motor Driver         | L298N               | Amazon        | 1     | R 120            | R 120       |
| 4           | DC Geared Motor      | TT 6V               | Amazon        | 2     | R 150            | R 300       |
| 5           | Wheel Encoder        | QTR-8RC             | RobotShop     | 2     | R 100            | R 200       |
| 6           | IMU Sensor           | MPU6050             | Amazon        | 1     | R 80             | R 80        |
| 7           | Steering Servo       | SG90                | Amazon        | 1     | R 150            | R 150       |
| 8           | Ultrasonic Sensor    | HC-SR04             | Amazon        | 1     | R 70             | R 70        |
| 9           | Emergency-Stop       | 22mm                | RS Components | 1     | R 90             | R 90        |
| 10          | Power Button         | PBS-110             | Amazon        | 1     | R 50             | R 50        |
| 11          | Rechargeable Battery | 7.4V 2200mAh        | Amazon        | 1     | R 350            | R 350       |
| 12          | Buck Converter       | LM2596              | Takealot      | 1     | R 60             | R 60        |
| 13          | Wheels               | 66mm                | Takealot      | 4     | R 60             | R 240       |
| 14          | 3D-Printed Chassis   | Custom              | Self-Design   | 1     | R 400            | R 400       |
| 15          | Wires/Connectors     | Jumper Kit          | Takealot      | 1 set | R 150            | R 150       |
| Grand Total |                      |                     |               |       |                  | R 4,310     |



## 5. Software Requirements

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- Arduino IDE
- ESP32 board support package
- C/C++ programming language
- LiDAR library / driver
- Servo motor library
- Encoder processing code
- IMU library
- Serial Monitor for debugging
- Git / GitHub (optional)



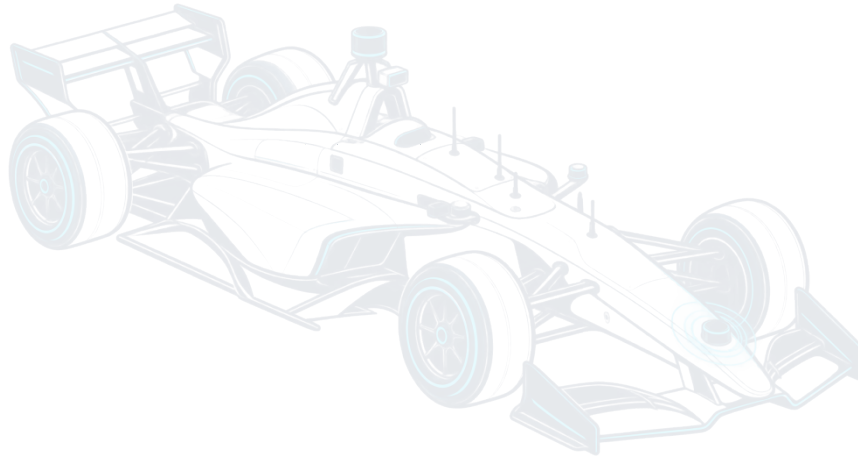


## CONTROL SYSTEM

# 6. System Architecture

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The ESP32-S3 acts as the main controller. It receives input from the 360° LiDAR (obstacle detection), wheel encoders (distance & speed), the IMU (orientation), and an ultrasonic sensor (backup). Based on this sensor data, it drives three outputs: the steering servo (angle control), the motor driver / DC motors (drive the wheels), and the emergency-stop system (safety).





## AUTONOMY CAPABILITIES

# 7. Main Features

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### Obstacle Detection

360° LiDAR continuously scans the surroundings and provides distance measurements.

### Distance Measurement

LiDAR measures distance to objects, while wheel encoders track how far the car has traveled.

### Automatic Steering

ESP32-S3 controls the steering servo according to the detected track direction.

### Speed Control

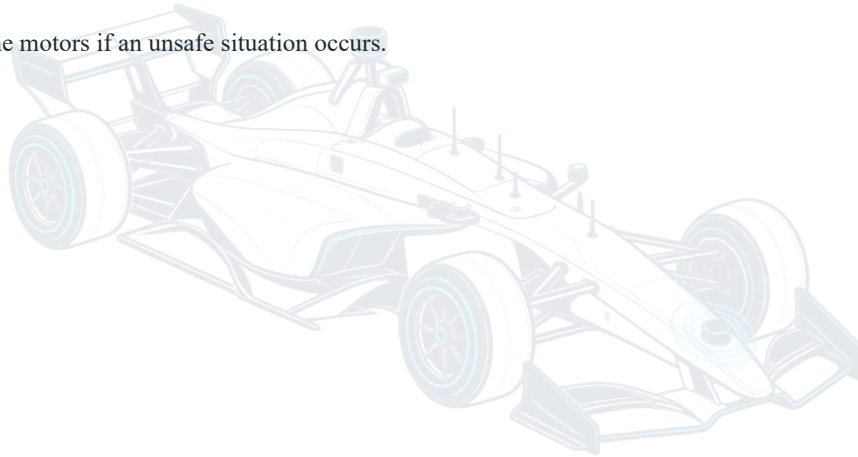
Motor speed is reduced before a turn or obstacle and increased after the turn.

### Collision Avoidance

Detects obstacles or other cars and takes action (slow down or change direction).

### Emergency Stop

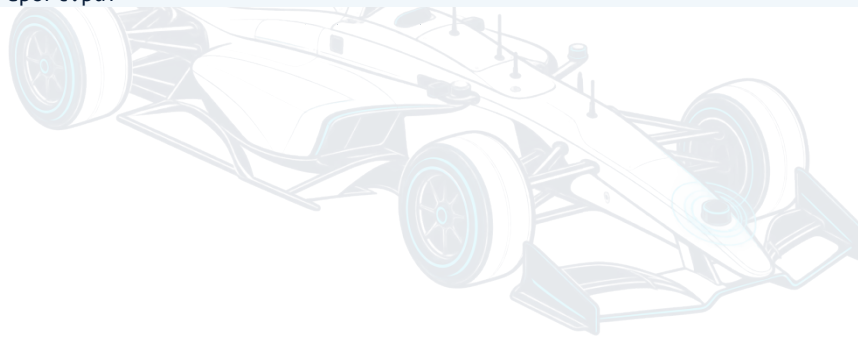
Immediately disables the motors if an unsafe situation occurs.





## 8. Project Structure

```
Autonomous-Racing-Car/  
├── src/  
│   └── main.cpp  
├── sensors/  
│   ├── lidar.cpp  
│   ├── encoder.cpp  
│   ├── imu.cpp  
│   └── ultrasonic.cpp  
├── motors/  
│   ├── motor_control.cpp  
│   └── steering.cpp  
├── control/  
│   ├── obstacle_avoidance.cpp  
│   ├── speed_control.cpp  
│   └── turn_control.cpp  
├── wiring/  
│   └── wiring_diagram.png  
├── CAD/  
│   └── chassis/  
└── documentation/  
    └── project_report.pdf
```



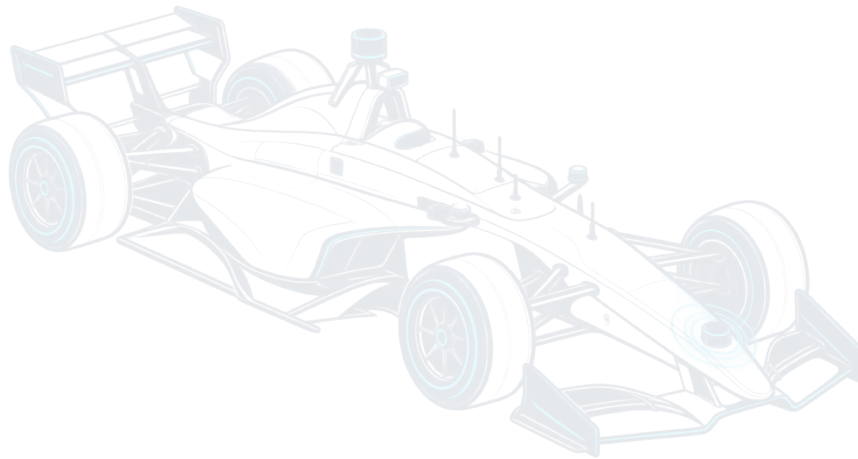


## SETUP WORKFLOW

# 9. Installation Guide

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1. Install Arduino IDE — Download and install the Arduino IDE.
2. Add ESP32 Support — Open File → Preferences → Add Board Manager URL: [https://dl.espressif.com/dl/package\\_esp32\\_index.json](https://dl.espressif.com/dl/package_esp32_index.json)
3. Install ESP32 Board Package — Tools → Board → Boards Manager → Search for ESP32 and install.
4. Select the Board — Tools → Board → ESP32-S3 Dev Module.
5. Connect the ESP32-S3 — Use a USB cable.
6. Install Required Libraries — LiDAR, Servo, Encoder, Motor control.
7. Upload the Program — Open src/main.cpp, compile and upload.





## CONTROL LOOP

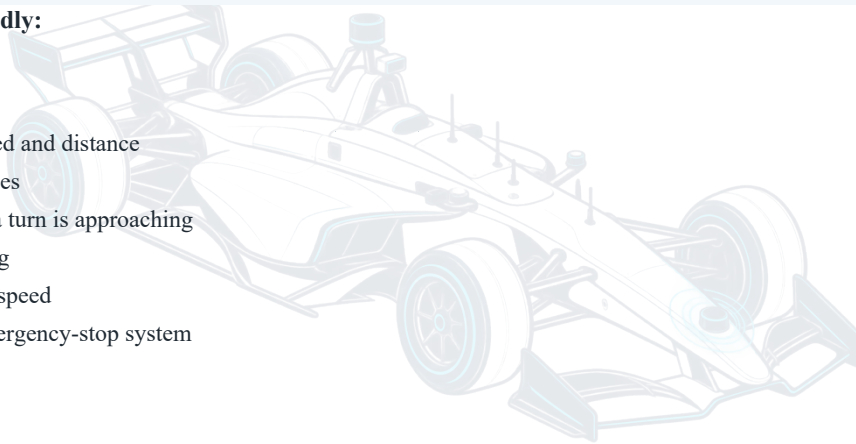
# 10. Code Instructions

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```
void loop() {  
  readLiDAR();  
  readEncoders();  
  readIMU();  
  
  calculateDistance();  
  calculateSpeed();  
  
  detectObstacles();  
  determineTurn();  
  
  controlSteering();  
  controlMotor();  
  
  checkEmergencyStop();  
}
```

The ESP32-S3 repeatedly:

8. Reads the sensors
9. Calculates distances
10. Calculates speed and distance
11. Detects obstacles
12. Determines if a turn is approaching
13. Adjusts steering
14. Adjusts motor speed
15. Checks the emergency-stop system



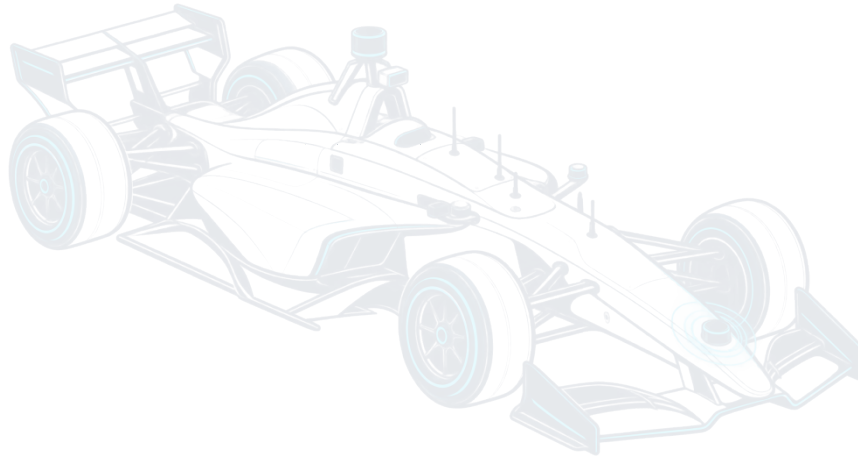


## TEST PROCEDURE

# 11. How to Run

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- Place the car on a safe test surface.
- Check all wiring.
- Verify LiDAR data.
- Check wheel encoders.
- Test steering servo.
- Test motor driver (wheels lifted).
- Test emergency-stop button.
- Upload the program.
- Start the system.
- Begin at low speed.
- Increase speed after verifying all systems.



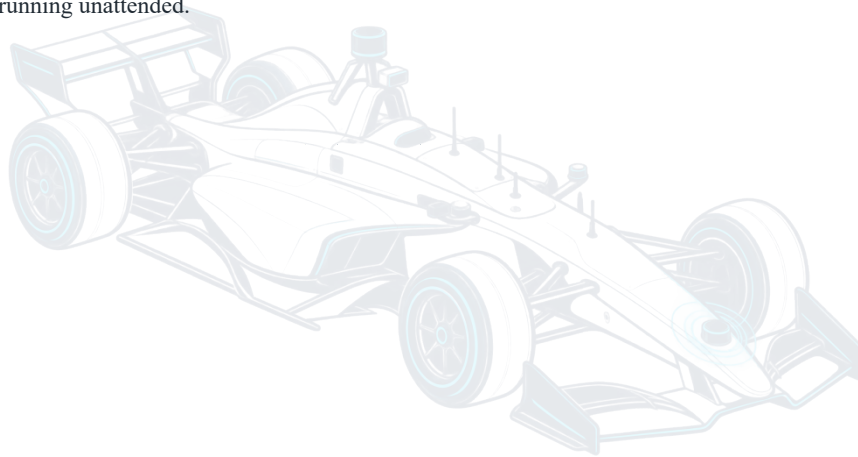


## SAFE OPERATION

# 12. Safety Notes

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- Always test at low speed first.
- Keep hands away from rotating wheels.
- Do not test near people, roads or hazards.
- Test the emergency-stop button.
- Ensure battery voltage is suitable.
- Do not connect motors to ESP32 GPIO pins directly.
- Use an appropriate motor driver.
- Make sure LiDAR and sensors are secure.
- Check for loose wires before every test.
- Disconnect the battery before changing wiring.
- Use proper battery protection.
- Never leave the car running unattended.



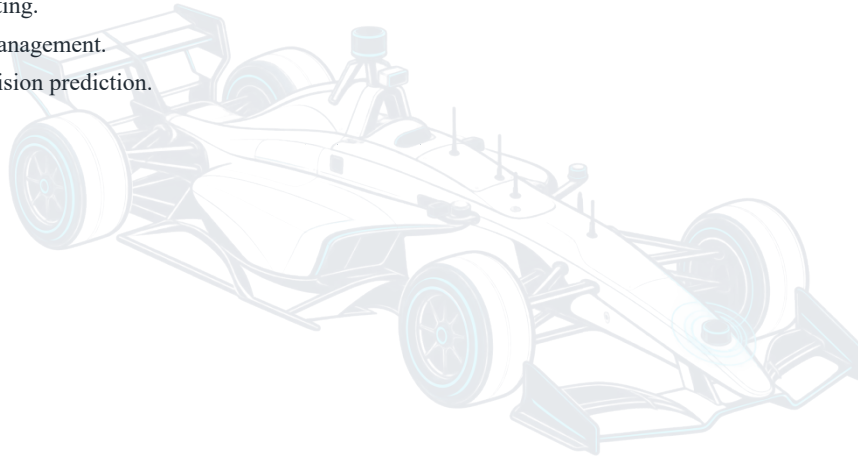


## NEXT LAP

# 13. Future Improvements

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- Higher-resolution LiDAR.
- Faster motor drivers.
- Better wheel encoders.
- GPS for outdoor navigation.
- Camera-based track recognition.
- Machine-learning obstacle detection.
- PID speed control.
- PID steering control.
- Automatic racing-line optimization.
- Wireless monitoring (phone/computer).
- Real-time telemetry.
- Automatic lap counting.
- Improved battery management.
- More advanced collision prediction.





## PROJECT TEAM

### 14. Authors

| Name | Role                    |
|------|-------------------------|
|      | Project Lead            |
|      | Hardware & Electronics  |
|      | Software & Programming  |
|      | Mechanical / CAD Design |
|      | Testing & Documentation |

Institution: \_\_\_\_\_

Course / Department: \_\_\_\_\_

Year: 2026

